

İZMİR UNIVERSITY OF ECONOMICS · FACULTY OF ENGINEERING · DEPARTMENT OF AEROSPACE ENGINEERING

FENG 498 — Engineering Capstone

Comparative Energy & CO₂ Analysis of Conventional, Hybrid and Electromagnetic Propulsion Systems for UAV Flight Stages

A Case Study on the Bayraktar TB2 MALE UAV

Kaoutar Ammara — ICE Model & Weather Decision Application

Hikmet Gökay Karakaya — Hybrid Propulsion Model

Elif Tosun — Electromagnetic Model & Comparison

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ICE Model & Weather App

Calibrated MATLAB baseline + pre-flight decision application

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Dual-source HESS configuration and results

06

Comparison & Conclusion

Unified KPI ranking and recommended architecture

Problem & Motivation

- Core Problem Statement: Tactical MALE UAV platforms face a significant operational paradox: their engines must be physically oversized to handle brief takeoff surges, causing severe part-load thermodynamic inefficiencies during extended loiter windows.
- Research Approach: Implemented a continuous, reduced-order, time-domain numerical mission simulation in MATLAB calibrated directly against empirical flight performance benchmarks.
- Scope Matrix: Evaluated three distinct propulsion configurations under identical longitudinal kinematics and fixed environmental conditions over a full 6.20-hour ISR profile.
- Primary Metric Focus: Prioritized cumulative source energy consumption, distance-normalized energy efficiency (km/MJ), and cumulative tailpipe carbon dioxide mass (CO₂).

Engineering question

Which propulsion architecture gives the best energy efficiency and lowest CO₂ for a TB2-class ISR mission — ICE, hybrid-electric, or electromagnetic?

700 kg

MTOW

100 hp

ICE baseline

6.2 h

Endurance

Sustainable Development Goals

SDG 7

Affordable & Clean Energy

Substituting ICE with hybrid / electromagnetic propulsion reduces fossil-fuel dependence in aerospace.

SDG 9

Industry, Innovation & Infrastructure

A calibrated, modular MATLAB framework + weather-aware decision app — a transferable mission-planning innovation.

SDG 13

Climate Action

The ICE model's 145.25 kg CO₂ / mission baseline anchors evaluation of cleaner configurations.

Project Overview

Three propulsion architectures evaluated on one airframe, one mission, one KPI set:

ICE

Conventional internal-combustion baseline (Rotax-class 100 hp). Highest endurance, simplest, highest direct CO₂.

Hybrid

Parallel ICE + electric motor with a 42 kWh battery. Cuts fuel in low-power phases; keeps range for peaks.

Electromagnetic

Electrically-assisted dual-source HESS (battery + supercapacitor). Smooths transients; limited by energy density.

Objective: Identify the most energy-efficient, lowest-CO₂ configuration across takeoff, climb, transit, loiter, descent and landing — under identical mission and atmospheric conditions.

TB2 ISR Mission Profile



Each phase has a distinct energy & emission signature. The same profile is applied to all three propulsion models, ensuring a fair, data-driven comparison. The 4-hour Loiter ISR phase dominates the energy budget (59.6 %).

PART I — AUTHOR: KAOUTAR AMMARA

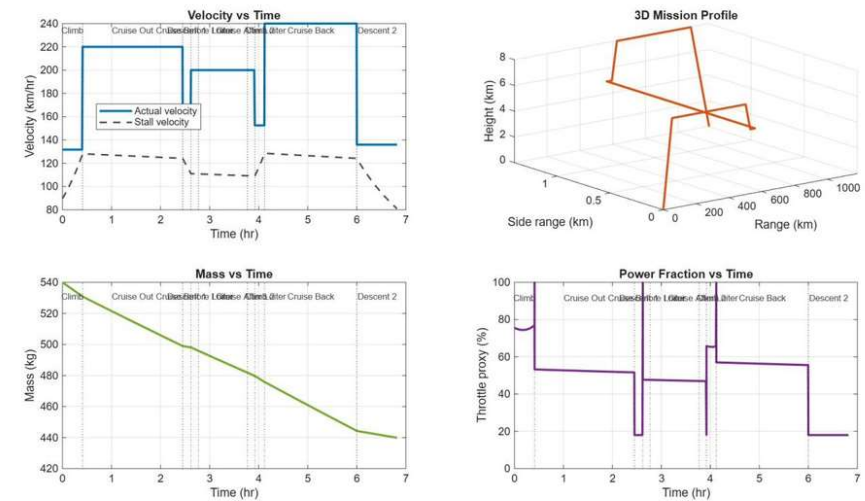
ICE Mission Simulation & Pre-Flight Weather-Gate App

Calibrated MATLAB point-mass mission model + Python (FastAPI + Streamlit) weather decision engine

Kaoutar Ammara — ICE Model & Weather Decision Application

Method & Calibration

- Reduced-order point-mass MATLAB simulation — forward Euler ($\Delta t = 1$ s), ISA atmosphere, parabolic drag polar.
- Built on the segmented method of Masood & Wei (2012); 9-segment benchmark as the calibration anchor.
- Uncalibrated burn 80.85 kg vs. published 101 kg ($\approx 20\%$ deficit) $\rightarrow$ BSFC scale factor 1.2493 $\rightarrow$ calibrated 100.16 kg.
- Calibrated fuel model then transferred to a TB2-class ISR mission profile.



Masood–Wei benchmark mission — velocity, 3D trajectory, mass & power-fraction (calibrated, variable atmosphere).

1.2493

BSFC scale factor

-0.83%

Calibrated error

9-seg

Benchmark

TB2-Class ISR Mission Results

45.97 kg

Total fuel used

145.25 kg

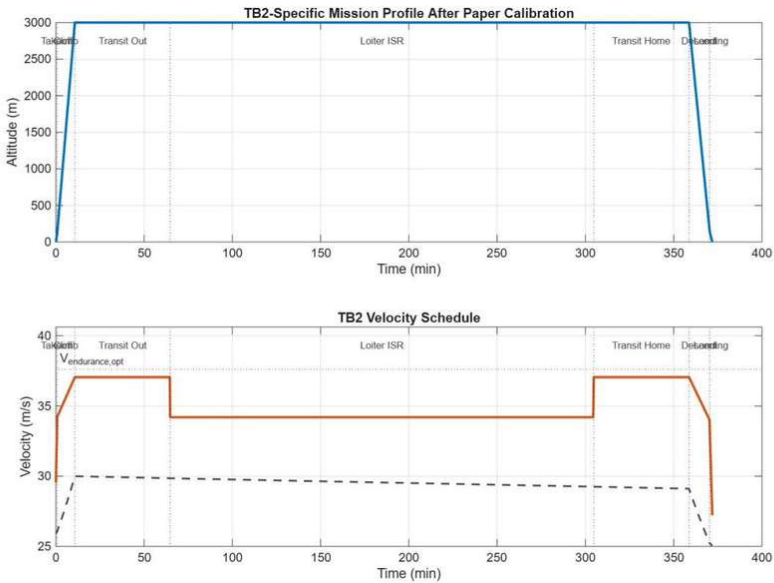
Tailpipe CO₂

782.59 km

Path length

6.20 h

Mission time



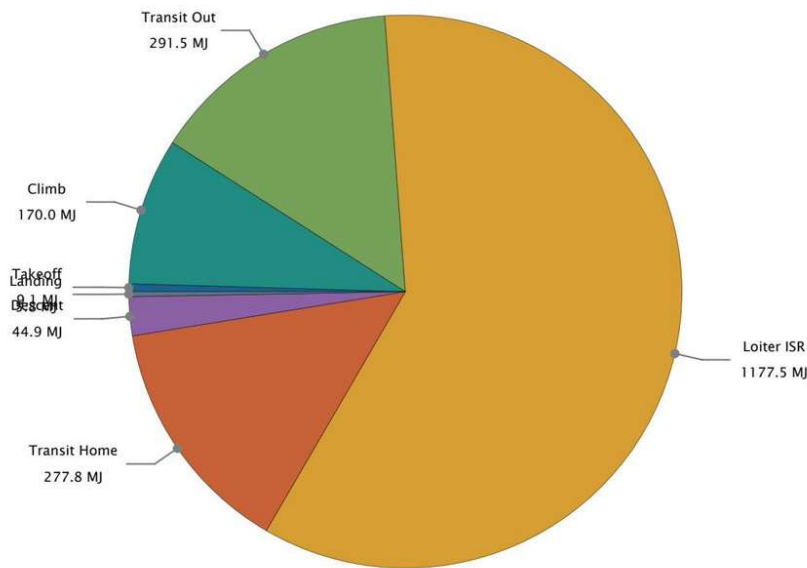
Altitude & velocity schedule across all seven flight phases.

Stage	Duration hr	Path km	Downrange km	Fuel kg	Energy MJ	Power kW	Thrust N
Takeoff	0.013	1.427	1.427	0.211	9.053	42.332	1063.000
Climb	0.167	21.184	21.184	3.954	170.030	59.326	1332.400
Transit Out	0.900	120.010	120.010	6.779	291.480	18.838	406.650
Loiter ISR	4.000	492.510	0.754	27.384	1177.500	17.128	400.620
Transit Home	0.900	120.010	119.880	6.459	277.750	17.945	387.380
Descent	0.194	24.706	24.706	1.043	44.865	13.423	0.000
Landing	0.025	2.741	2.741	0.134	5.768	13.423	3.088

Speeds set by optimization: transit at min fuel/km, loiter at min fuel/hour (V≈34 m/s). Per-stage duration, fuel, energy, power & thrust shown above.

Mission Energy Breakdown

TB2 Energy Share by Stage

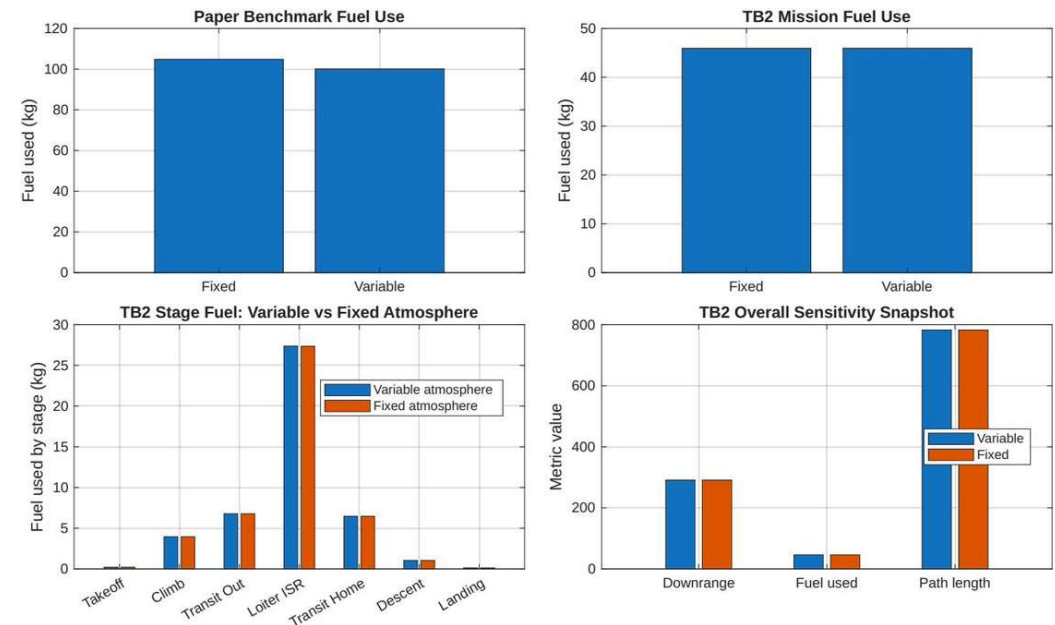


Loiter ISR = 1,177.5 MJ → 59.6% of total mission energy

- Total mission energy: 1,976.5 MJ across the seven flight phases.
- The 4-hour endurance window dominates — far above takeoff, climb and transit combined.
- Endurance-phase efficiency is therefore the highest-leverage target for hybrid / electric alternatives.
- This is the key benchmark the hybrid and EM models are measured against.

Atmosphere Sensitivity Study

- One-factor study: variable-atmosphere vs. fixed-atmosphere simulation.
- TB2 ISR mission at 3,000 m: all key outputs differ by < 0.15 %.
- Fuel -0.063 %, CO₂ -0.028 % — well below the 2 % acceptance threshold.
- Fixed atmosphere is justified for the three-model comparison (vs. +4.70 % error on the high-altitude paper mission).



-0.063%

Fuel difference

-0.028%

CO₂ difference

< 0.15%

Max key output

Paper benchmark (large fixed-atmosphere error) vs. TB2 mission (negligible error) — fuel, stage fuel & overall snapshot.

Cost Analysis vs. Rotax Reference

Parameter	This Work (ICE)	Rotax 914 Ref.
BSFC base (kg/kW·h)	0.320	~0.300–0.340
Calibrated BSFC (kg/kW·h)	0.3998	~0.320 (cruise)
Mission fuel used (kg)	45.97	101.0 (paper)
Fuel energy (MJ)	1,976.5	4,306.8 (paper)
CO ₂ per mission (kg)	145.25	316.5 (paper)
Fuel cost (~\$1.50/kg)	~\$69	~\$152

- Calibrated BSFC of 0.40 kg/kW·h is consistent with mid-throttle Rotax 914 cruise operation.
- TB2 mission burns 45.97 kg vs. the paper mission's 101 kg — a different, shorter mission, not a like-for-like comparison.
- Indicative fuel cost ≈ \$69 per mission establishes the ICE economic baseline for the hybrid / EM comparison.

Weather-Gate Decision Application

- Python pre-flight mission-readiness app built alongside the MATLAB simulation — FastAPI backend + Streamlit dashboard.
- Pulls live Open-Meteo forecast + air-quality data for any location, date and time.
- Evaluates 10 weather constraints from the TB2 envelope; overall verdict = most-restrictive state.
- Returns optimal altitude, recommended TAS, fuel estimate and latest safe return time.

The screenshot shows the 'TB2 Mission Weather Gate' dashboard. At the top right, it displays '46.3 m/s' and '4877 m'. The main section is divided into 'Mission Input' and 'Decision'. The 'Mission Input' section contains several sliders and text inputs for location, date, time, duration, altitude, heading, and return response. A red 'Evaluate Mission' button is at the bottom.

Streamlit dashboard — main interface



10 weather constraints — overall verdict = most-restrictive state

MISSION APPROVED

CAUTION

MISSION REJECTED

Three Decision Cases

Ankara

MISSION APPROVED · 100 / 100

her Gate

46.3 m/s

Operational altitude

4877 m

Decision

Mission Approved | mission readiness score 100/100

Ankara | Lat 39.929N, Lon 32.834E

Suggested altitude

4000 m

Suggested TAS

40.9 m/s

Estimated fuel

12.5 kg

Latest return

100 min

Constraint Checks

Check	Status	Measured	Limit	Units	Message
Steady wind	PASS	3.49	16.205	m/s	Steady wind is pass
Wind gust	PASS	5.4	20.835	m/s	Wind gust is pass
Crosswind component	PASS	0.997	10.887	m/s	Crosswind component is pass
Precipitation	PASS	0.0	2.0	mm/h	Precipitation is pass
Snowfall	PASS	0.0	0.5	cm/h	Snowfall is pass
Visibility	PASS	4000.0	3000.0	m	Visibility is pass
Ambient temperature	PASS	-21.0 to -25.2	-20 to -45	deg C	Temperature is pass
WMO weather code	PASS	1.0	No fog, freezing rain, heavy rain		Open Window WMO code
Humidity and fog tendency	PASS	48% RH, 420	90% RH with reduced visibility		High humidity is not
PM10 or smoke/dust proxy	PASS	25.4	150.0	µg/m³	Air quality data is not

Weather Window

time	temperature_2m	relative_humidity_2m	apparent_temperature	precipitation_probabil
2026-06-12T00:00	26.7	97	32	
2026-06-12T01:00	26.4	88	31.4	

All 10 constraints PASS.

Hong Kong

CAUTION · 82 / 100

Decision

Proceed with Caution | mission readiness score 82/100

Hong Kong | Lat 22.2793N, Lon 114.1741E

Suggested altitude

3000 m

Suggested TAS

36.6 m/s

Estimated fuel

11.0 kg

Latest return

100 min

Constraint Checks

Check	Status	Measured	Limit	Units	Message
Steady wind	PASS	2.4	16.205	m/s	Steady wind is pass
Wind gust	PASS	4.9	20.835	m/s	Wind gust is pass
Crosswind component	PASS	0.107	10.887	m/s	Crosswind component is pass
Precipitation	PASS	0.1	2.0	mm/h	Precipitation is pass
Snowfall	PASS	0.0	0.5	cm/h	Snowfall is pass
Visibility	WATCH	1940.0	3000.0	m	Visibility is watch
Ambient temperature	PASS	-25.4 to -26.7	-20 to -45	deg C	Temperature is pass
WMO weather code	WATCH	5.0	No fog, freezing rain, heavy rain		Open Window WMO code
Humidity and fog tendency	PASS	88% RH, 294	90% RH with reduced visibility		High humidity is not
PM10 or smoke/dust proxy	PASS	29.7	150.0	µg/m³	Air quality data is not

Weather Window

time	temperature_2m	relative_humidity_2m	apparent_temperature	precipitation_probabil
2026-06-12T00:00	26.7	97	32	
2026-06-12T01:00	26.4	88	31.4	

Visibility & WMO code on WATCH.

New Delhi

MISSION REJECTED · 16 / 100

Decision

Mission Rejected | mission readiness score 16/100

New Delhi | Lat 28.6234N, Lon 77.2149E

Suggested altitude

4000 m

Suggested TAS

40.6 m/s

Estimated fuel

12.4 kg

Latest return

0 min

Constraint Checks

Check	Status	Measured	Limit	Units	Message
Steady wind	PASS	3.11	16.205	m/s	Steady wind is pass
Wind gust	PASS	6.7	20.835	m/s	Wind gust is pass
Crosswind component	PASS	1.034	10.887	m/s	Crosswind component is pass
Precipitation	WATCH	1.2	2.0	mm/h	Precipitation is watch
Snowfall	PASS	0.0	0.5	cm/h	Snowfall is pass
Visibility	FAIL	2960.0	3000.0	m	Visibility is fail
Ambient temperature	PASS	-22.7 to -28.6	-20 to -45	deg C	Temperature is pass
WMO weather code	WATCH	55.0	No fog, freezing rain, heavy rain		Open Window WMO code
Humidity and fog tendency	PASS	87% RH, 296	90% RH with reduced visibility		High humidity is not
PM10 or smoke/dust proxy	FAIL	284.8	150.0	µg/m³	Air quality data is not

Weather Window

time	temperature_2m	relative_humidity_2m	apparent_temperature	precipitation_probabil
2026-06-12T00:00	28.6	79	33	
2026-06-12T01:00	25.7	87	29.8	
2026-06-12T02:00	26.3	85	31.3	

Visibility FAIL (2,960 m); PM₁₀ FAIL (284 µg/m³).

Comparative UAV Propulsion Analysis — Bayraktar TB2

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PART II — AUTHOR: HIKMET GÖKAY KARAKAYA

Parallel Hybrid-Electric Propulsion Model

Actively-managed ICE + electric motor with a 120 kg / 42 kWh semi-solid-state battery and a rules-based 50/50 power split

Hikmet Gökay Karakaya — Hybrid Propulsion Model

Architecture & Power-Split Strategy

- Parallel config: 100 hp ICE + 100 hp (74.57 kW) brushless-DC motor on a common propeller shaft.
- 120 kg semi-solid-state battery, 350 Wh/kg → 42.0 kWh capacity; usable window 100 % → 10 % SoC.
- Rules-based power split adapts by phase to protect reserves and maximise carbon displacement.
- Engine locked into its narrow optimal BSFC band (285 g/kW·h ≈ 29.4 % thermal efficiency); electric chain at 80 %.
- Parallel Hybrid System: Both the ICE and the DC motor are connected directly to the shaft. A computer-controlled clutch engages or disengages either motor so the shaft can be driven by the ICE alone, the DC motor alone, or both simultaneously for maximum acceleration.

Takeoff / Climb

ICE only ($\alpha=1.0$) — battery held at 100 % SoC

Transit / Loiter

50 / 50 split — ICE in low-burn band + electric torque assist

Descent

ICE idle — near-zero fuel (0.001 kg)

Landing

All-electric ($\beta=1.0$) — silent, zero-emission approach

Results — Fuel, CO₂ & State-of-Charge

27.77 kg

Fuel (-39.58 %)

87.77 kg

CO₂ (-39.58 %)

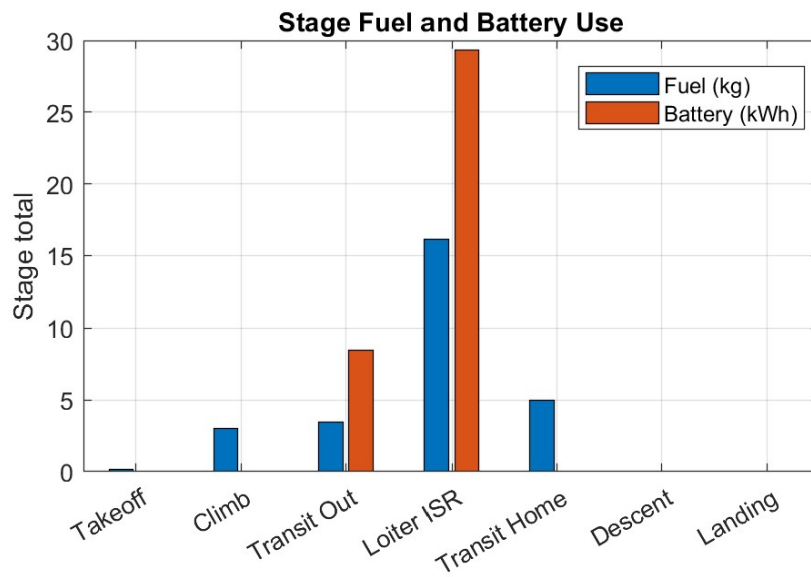
10.00 %

Final battery SoC

267.5 min

SoC floor reached → ICE fallback

Hybrid Fuel vs Battery Usage by Stage



- 50/50 split during transit & loiter shifts half the propulsive load to the electric drive.
- Battery drains steadily (mean 146.5 A in loiter) to its 10 % floor at 267.5 min; ICE fallback resumes seamlessly.
- 39.58 % cut in both fuel and CO₂ vs. the calibrated ICE baseline — 57.5 kg CO₂ displaced per mission.

Cost Analysys for Hybrid Case

Direct Cost Element Parameter	Hybrid Per-Mission Cost	Engineering System Sizing Basis & Notes
Fuel Consumed Cost	\$41.65	Based on the optimized 27.77 kg fuel burn at an indicative \$1.50/kg.
Electricity Consumed Cost	\$5.67	Cost to replenish 37.80 kWh from the ground grid at a \$0.15/kWh rate.
Battery Pack Depreciation Cost	\$59.92	Derived from a \$14,500 initial CAPEX divided by an aerospace limit of 242 flight cycles.
74.57 kW Brushless DC Motor Cost	\$49.59	Derived from a \$12,000 asset cost amortized over 242 missions (1,500 flight hours).
TOTAL DIRECT OPERATIONAL COST	\$156.83	Combined per-flight energy and high-value drivetrain asset depreciation cost.

- Direct Energy Efficiency: The 50/50 parallel power-split shifts half the propulsive load to the electric drive, dropping fuel costs to just \$41.65 per mission.
- Low-Cost Electricity: Recharging the battery pack back from its 10% safety floor to full capacity costs only \$5.67 per flight.
- Amortized Capital Asset Costs: High-value hardware components are completely amortized over a 5-year, 242-mission lifecycle framework to accurately map component depreciation.
- Total Mission Financial Impact: Combining immediate energy consumption with predictable hardware wear results in an optimized total direct cost of \$156.83 per flight profile.

PART III — AUTHOR: ELIF TOSUN

Electrically-Assisted (EM) Propulsion Model

Dual-source Hybrid Energy Storage System (HESS): lithium battery + supercapacitor matrix with low-pass frequency-separation control

Elif Tosun — Electromagnetic Model & Comparison

Dual-Source HESS Architecture

- Electrically-assisted core: primary ICE + a dual-source Hybrid Energy Storage System (battery + supercapacitor).
- A time-domain low-pass frequency-separation filter routes power dynamically — not a fixed rule-based split.
- High-frequency torque surges → supercapacitor (peak-shaving); smooth low-frequency loads → lithium battery.
- Protects primary cells from current ripple, voltage drops and thermal stress during takeoff & climb.

Takeoff / Climb

Supercapacitor shaves transient peaks; battery smooths background load — ICE ramps linearly.

Transit / Loiter

Steady aerodynamic load → supercapacitor idle; battery sustains low-frequency assist.

Battery exhaustion

At 10 % SoC the assist disconnects; ICE assumes 100 % of remaining loiter workload.

Results — Engine-Dominant Outcome

45.90 kg

Total fuel used

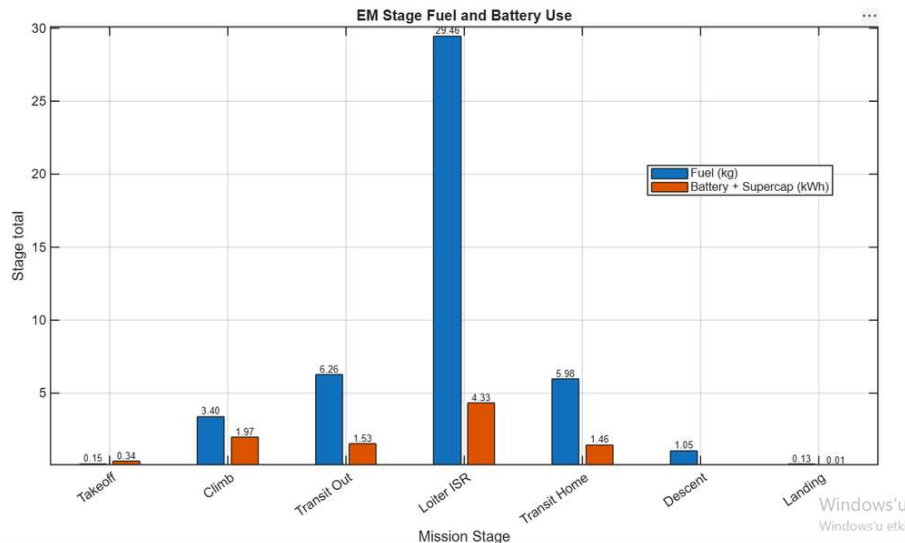
145.00 kgTailpipe CO₂**0.42 km/MJ**

Mission efficiency

2,020 MJ

Source energy

Fuel burn vs. ICE baseline



- Transient electrical reserves deplete early in the 4-hour loiter; the ICE then runs unassisted for most of the mission.
- Fuel (45.90 kg) and CO₂ (145.00 kg) track the ICE baseline almost exactly — negligible mission-level benefit.
- Unused storage cells add dead-weight, pushing source energy (2,020 MJ) slightly above the ICE baseline.
- HESS excels at component protection & torque smoothing, not long-endurance carbon reduction.
- EM-assisted propulsion improves transient response but does not significantly reduce mission-level fuel and CO₂ because the 4-hour loiter phase remains ICE-dominant.

Cost Analysis for EM Case

Direct Cost Element Parameter	EM Per-Mission Cost	Engineering System Sizing Basis & Notes
Fuel Consumed Cost	\$68.85	Based on the EM-assisted 45.90 kg fuel burn at an indicative \$1.50/kg fuel cost.
Electricity Consumed Cost	\$0.95	Minor recharge cost for the limited battery-supercapacitor assist system after one mission.
Primary Lithium Battery Depreciation Cost	\$43.39	Derived from a \$10,500 battery pack CAPEX divided by 242 mission cycles.
Supercapacitor Matrix Depreciation Cost	\$24.79	Derived from a \$6,000 transient supercapacitor matrix divided by 242 mission cycles.
35 kW Electric-Assist Motor & Controller Cost	\$49.59	Derived from a \$12,000 assist motor/controller asset amortized over 242 missions.
TOTAL DIRECT OPERATIONAL COST	\$187.57	Combined fuel, electricity, battery, supercapacitor, and electric-assist drivetrain depreciation cost.

Fuel-Dominant Cost Structure:

- The EM-assisted case still depends mainly on the ICE during the long loiter phase, so fuel remains the largest direct cost item.

Low Electricity Cost:

- Electrical recharge cost is very small because the battery and supercapacitor are used mainly for transient assist, not full-mission propulsion.

Added HESS Depreciation:

- Unlike the ICE case, the EM-assisted system includes both a lithium battery pack and a supercapacitor matrix, increasing amortized hardware cost.

Cost-Performance Trade-Off:

- Although the EM system improves takeoff support and peak-load buffering, its total direct cost is higher than the hybrid case because it does not significantly reduce fuel burn.

PART IV — COMPARISON & CONCLUSION

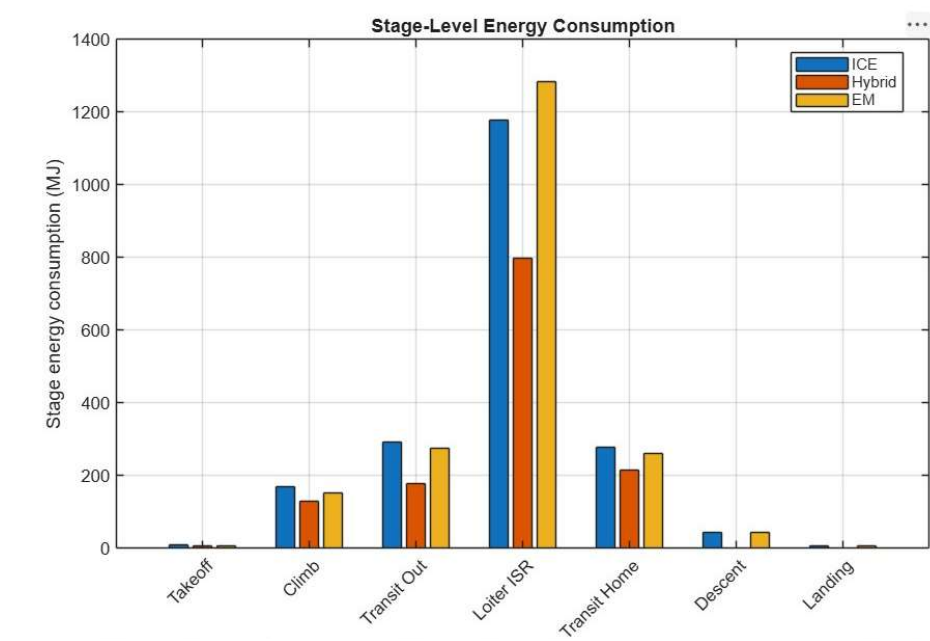
Unified KPI Comparison & Recommendation

All three architectures evaluated on one mission, one atmosphere, one KPI set — source energy, efficiency and tailpipe CO₂

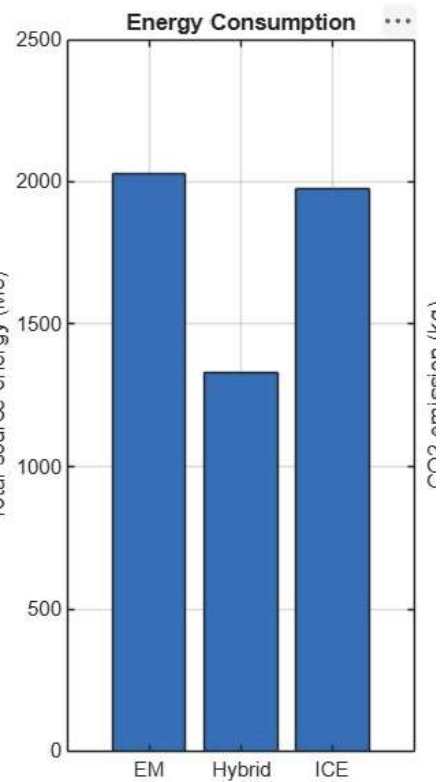
Joint analysis — Elif Tosun & team

Source Energy & Efficiency

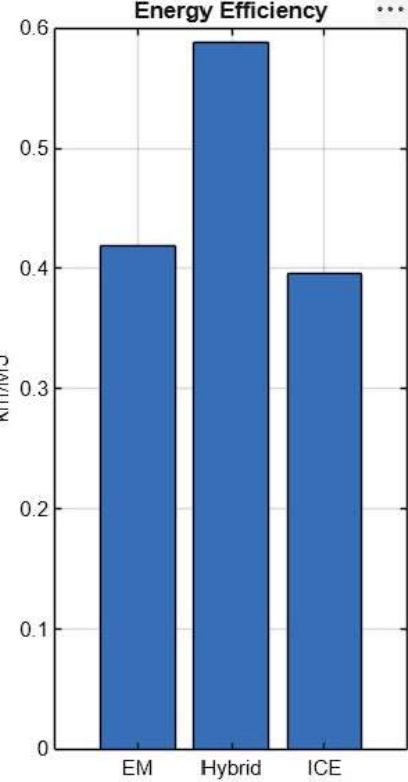
Stage energy Consumption (MJ)



Cumulative source energy (MJ)



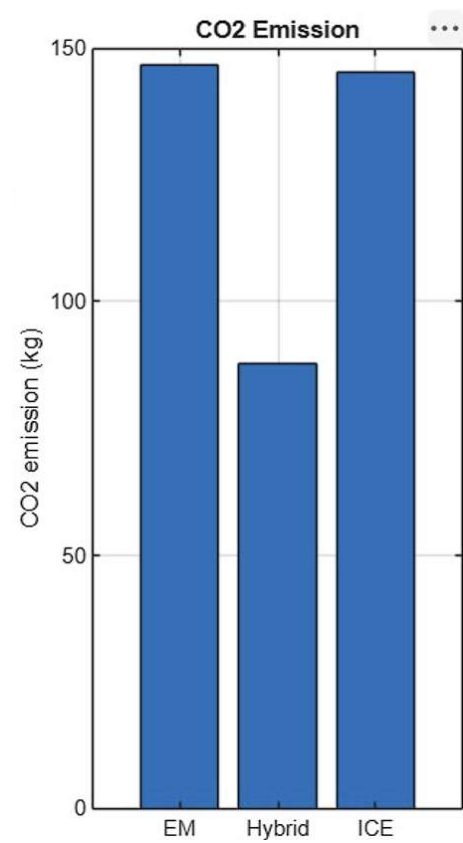
Mission efficiency (km / MJ)



Hybrid: 1,350 MJ (–31.7 % energy) and 0.59 km/MJ (+47.5 % efficiency) — best of the three. EM sits slightly above the ICE baseline on energy.

CO₂ & Unified KPI Matrix

Tailpipe CO₂ (kg)



KPI	ICE	Hybrid	EM
Mission time (h)	6.20	6.20	6.20
Distance (km)	782.59	782.59	782.59
Source energy (MJ)	1,976.5	1,350.0	2,020.0
Efficiency (km/MJ)	0.40	0.59	0.42
Tailpipe CO ₂ (kg)	145.25	87.77	145.00
Ranking	Baseline	1st	2nd

Hybrid wins on every KPI it can — lowest energy, highest efficiency, lowest CO₂.

CONCLUSION

Recommended Architecture

The Parallel Hybrid-Electric model is recommended for the TB2 class — 39.58 % lower fuel & CO₂ and 0.59 km/MJ (+47.5 %) efficiency.

- ICE baseline (calibrated to -0.83 % vs. benchmark): 45.97 kg fuel, 145.25 kg CO₂, 782.59 km in 6.20 h; Loiter ISR = 59.6 % of mission energy.
- Hybrid: 50/50 split locks the ICE in its optimal band → 27.77 kg fuel / 87.77 kg CO₂, plus a silent all-electric landing.
- EM-assisted: dual-source HESS protects components and smooths transients, but stays engine-dominant (45.90 kg / 145.00 kg) — minimal mission-level carbon benefit.
- Because the 4-hour loiter dominates the energy budget, only a configuration that optimises steady-state cruise achieves real gains — exactly what the hybrid does.

CONCLUSION

Future Research & Recommendations

1

Predictive supervisory control

Replace fixed rule-based splits with Model Predictive Control using live telemetry + Open-Meteo weather to optimise energy splits under varying winds.

2

Multi-variable mass optimization

Map dead-weight penalties of battery thermal-management systems and supercapacitor sizing against airframe downrange limits.

3

Expanded lifecycle boundaries

Move beyond tailpipe CO₂ to well-to-wake emissions, including grid-charging carbon intensity and non-CO₂ radiative forcing.

Thank You

Questions & Discussion

Comparative Energy & CO₂ Analysis of ICE, Hybrid and Electromagnetic Propulsion — Bayraktar TB2 MALE UAV

Kaoutar Ammara · Hikmet Gökay Karakaya · Elif Tosun | Supervisor: Prof. Dr. Daa Gadelmavla